AA244B: Advanced Plasma Physics and Engineering
Spring 2023, Stanford University

Project

Due: June 14th (Wed) 11:59 pm PT

In this project, you will construct your own particle-in-cell simulation to visualize two-stream instability and Landau damping. The problems are designed such that the PIC simulations can be built using Matlab and the computational wall time is about a few minutes on a regular desktop computer. However, feel free to use some more advanced coding languages (e.g., C/C++, Fortran)

Governing equations

The governing equations are given by the equations of motion and the Poisson equation for electrostatic cases. Here, we focus on one-dimensional problems. First, let us introduce normalized variables.

$$\tilde{x} = \frac{x}{\lambda_D}; \tilde{v} = \frac{v}{v_{th,e}}; \tilde{t} = \omega_{pe} t = \frac{v_{th,e}}{\lambda_D} t;\ \tilde{E} = \frac{e\lambda_D}{m v_{th,e}^2} E = \frac{E}{\left(\frac{T_{eV}}{\lambda_D}\right)};\ \tilde{B} = \frac{eB}{m\omega_{pe}}; \tilde{n} = \frac{n}{n_0}$$

where $\lambda_D = \left(\frac{\epsilon_0 k_B T_e}{e^2 n_0}\right)^{1/2} = \left(\frac{\epsilon_0 T_{eV}}{e n_0}\right)^{1/2}$, $v_{th,e} = \left(\frac{k_B T_e}{m_e}\right)^{1/2}$, and $\omega_{pe} = \left(\frac{e^2 n_0}{m_e \epsilon_0}\right)^{1/2}$.

The equations of motion can be written as

$$\frac{d\tilde{x}}{d\tilde{t}} = \tilde{v},$$

$$\frac{d\tilde{v}}{d\tilde{t}} = \pm\left(\tilde{E} + \tilde{v} \times \tilde{B}\right),$$

where $+$ is for ions and $-$ is for electrons.

The Gauss's law can be given

$$\frac{d\tilde{E}}{d\tilde{x}} = \tilde{\rho} = \tilde{n}_i - \tilde{n}_e.$$

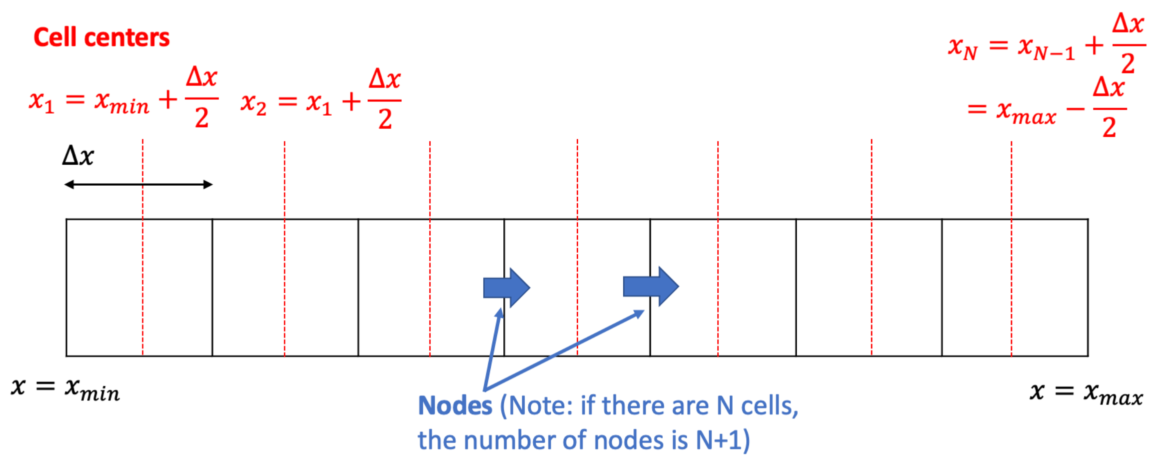
Note that if $\tilde{\phi} = \phi / T_{eV}$ is introduced, the Poisson equation can be given as

$$\frac{d^2\tilde{\phi}}{d\tilde{x}^2} = -(\tilde{n}_i - \tilde{n}_e).$$

Further if we consider, immobile ions, then $\tilde{n}_i = 1$.

Simulation procedure

First thing you need to decide is whether you put the variables on cell centers or nodes. See figure below. The following is my suggestion how to proceed, but you can implement the code with different configurations.



1. Initialize parameters (domain size, grid size, time step, total time to run, number of macroparticles per cell or total number of macroparticles in the system, output frequency, etc).

2. Initialize cell/grid arrays on cell center (x, ϕ, ρ, n_i, n_e). You can simultaneously allocate the variables for output (e.g., $\vec{u}, T, \epsilon$).

3. Initialize field quantities on nodes (E). This way the interpolation to the particles can be done easily. Note that if you have N cells, then you will have N+1 nodes.

4. Initialize particles. Set the total number of macroparticles in the system. Each macroparticle should have position, velocity, particle weight, charge, and mass. Here, in this test case, we will focus on immobile ion cases, thus the species of interest would be electrons. Note that particle weight = density / the number of macroparticles in the cell. So, if you decide to initialize the density as $\tilde{n} = 1$ and if you have $N_{ppc} = 100$, each macroparticle has a weight of 0.01.

5. Start iteration

   a. Gather the density (on the cell centers) from the particles
   b. Calculate the charge density $\tilde{\rho} = \tilde{n}_i - \tilde{n}_e$
   c. Solve the Poisson equation to obtain $\tilde{\phi}$. Note that if your charge density is on cell centers, the potential will also be on cell centers.
   d. Calculate electric field from $\tilde{E} = -d\tilde{\phi}/d\tilde{x}$. Here, if the potential is on cell centers, $\tilde{E}_{i-1/2} = -(\tilde{\phi}_i - \tilde{\phi}_{i-1})/\Delta x$ gives second-order accuracy.
   e. Interpolate the electric field exerted on each macroparticle from the nodes.
   f. Move the particle position according to the updated velocity.
   g. Save data (e.g., bulk velocity, total energy, electrostatic energy) for output, if needed

6. Output and/or visualize data

Numerical scheme

- Use leap frog scheme for position and velocity update

$$\frac{\tilde{v}^{n+1/2} - \tilde{v}^{n-1/2}}{\Delta\tilde{t}} = \tilde{E}^n(\tilde{x}^n),$$

$$\frac{\tilde{x}^{n+1} - \tilde{x}^n}{\Delta\tilde{t}} = \tilde{v}^{n+1/2}.$$

  Note the velocity update can also be written as

$$\frac{\tilde{v}^n - \tilde{v}^{n-1/2}}{\Delta\tilde{t}/2} = \frac{\tilde{v}^{n+1/2} - \tilde{v}^n}{\Delta\tilde{t}/2} = \tilde{E}^n(\tilde{x}^n),$$

  This half time step consideration is important when you want to visualize position and velocity, strictly speaking.

  In addition, note that if $\tilde{x}^{n+1}$ (update position) is outside of the domain, then you have to do something about it. For instance, if the charge particles are neutralized at the wall, then that macroparticle will need to be deleted. If periodic boundary condition is used, then you need to put it back into the domain.

- Use linear interpolation (i) for the charge deposition from the macroparticles to the cell centers and (ii) for the electric field from the nodes to the macroparticles. For instance, the latter can be written as

$$\tilde{E}(\tilde{x}) = \frac{\left(\tilde{x}_{i+1/2} - \tilde{x}\right)}{\Delta\tilde{x}}\tilde{E}_{i-1/2} + \frac{\left(\tilde{x} - \tilde{x}_{i-1/2}\right)}{\Delta\tilde{x}}\tilde{E}_{i+1/2}$$

- Poisson equation in periodic boundary condition can be solved using FFT (fast Fourier transform). Verify that this FFT works with an analytic theory (e.g., assuming sinusoidal profile for the right hand side of the Poisson equation). The procedure is as follows.

  1. Take the mean $\langle\tilde{\rho}\rangle = \frac{1}{x_{max}-x_{min}}\int_{x_{min}}^{x_{max}}\tilde{\rho}dx$, and calculate $\tilde{\rho}' = \tilde{\rho} - \langle\tilde{\rho}\rangle$
  2. Conduct FFT of $\tilde{\rho}'$. Let's call this R.
  3. Create a new array

$$k_j = \begin{cases} \frac{\pi}{\Delta x}\frac{j-1}{N/2} & for\ 1 \le j \le \frac{N}{2} \\ \frac{\pi}{\Delta x}\frac{j-1}{N/2} - 2\frac{\pi}{\Delta x} & for\ \frac{N}{2} < j \le N \end{cases}$$

  4. Calculate

$$Y_j = +\frac{R_j}{\kappa_j^2},$$

     where

$$\kappa_j = \frac{\sin\left(\frac{k_j\Delta x}{2}\right)}{\frac{\Delta x}{2}}.$$

Note that j = 1 would give you a singularity. Hence, consider a small value for $k_{j=1}$, e.g., 10^{-8}.

5. Conduct inverse FFT (e.g. ifft in matlab) of Y. Let us call this $\hat{Y}$.
6. $\tilde{\phi}_j$ is the real part of $\hat{Y}$ (may need to take the mean of $\tilde{\phi}$ to make the spatial average to be zero)

- To randomize the particle position $\tilde{x} = R(\tilde{x}_{max} - \tilde{x}_{min}) + \tilde{x}_{min}$, where $0 \le R < 1$ is a random number.

- To initiate a sinusoidal density perturbation, you can calculate $\delta\tilde{x} = \frac{\delta\tilde{n}}{\tilde{k}}\sin(\tilde{k}\tilde{x})$, where $\tilde{k}$ is the wave number of the perturbation that you want to apply and $\delta\tilde{n}$ is the amplitude of the density perturbation ($\delta\tilde{n} < \tilde{n}_0 = 1$). Correct the particle position, $\tilde{x} = \tilde{x} + \delta\tilde{x}$.

- To initiate a Maxwellian VDF in 1D, you can use acceptance-rejection method or $\tilde{v} = \sqrt{-\ln(R_1)}\cos(2\pi R_2)$, where R_1 and R_2 are two different random numbers.

<u>Test cases to study</u>

Set the domain size $\tilde{L} = 2\pi$ and start from the number of cells $N_c = 32$. Timestep can be $\Delta\tilde{t} = 0.04$. The total time to run $\tilde{T} = 30$.

1. **Two-stream instability**: set one electron beam with $\tilde{v} = 0.5$ and another electron beam with $\tilde{v} = -0.5$. Set the number of macroparticles to be $N_{ppc} = 50$. Observe if you get instability. What happens if you change the velocity of the counterpropagating beams? To check the instability growth and saturation, we typically monitor the electrostatic energy $\tilde{\epsilon}_{wave} = \frac{1}{2}\int|\tilde{E}|^2 d\tilde{x}$. The slope of the linear growth of the electrostatic energy should be scale as 2γ since γ is the growth rate of the electric field.

2. **Landau damping**: initialize a stationary Maxwellian distribution function. Apply a sinusoidal density perturbation, $\delta n = 0.3$. Test your code with $N_{ppc} = 1000$ to get good statistics. Assume a single mode, $\tilde{k} = 1$. The initial density fluctuation should give rise to electric field fluctuation, i.e., a finite electrostatic energy. Observe if you get the electrostatic energy to decrease in time. What happens if you change the amplitude of the initial number density fluctuation?

<u>What to investigate and put in the report</u>

Plot a few time snapshots of the phase space for the report. It is useful to create a movie to understand dynamic phenomena, but unfortunately you cannot attach a movie to a report. If possible, one can bin the particles into discretized phase space and construct the velocity distribution function, $f(x, v)$. Plot the corresponding electric field and electron number density at the selected times.

Plot the time history of the electrostatic energy and total kinetic energy in log scale. Apply a curve fit to the linear instability phase and estimate the growth/damping rate. Discuss whether the growth/damping rate is as expected from the analytic theory.

For both cases, study the particle convergence, namely, try a few (3-5) more runs with different number of macroparticles (for instance, x0.5, x2, x4, x10).